

1 Deep sets

Consider an exchangeable function given by

$$f(x_1, x_2, \dots, x_M) = f(x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(M)})$$

for any permutation $\pi(\cdot)$, and suppose we are interested in a classification or regression problem to estimate the function $f : \mathcal{X}^M \rightarrow \mathcal{Y}$. For a set of size n , there are $n!$ permutations, so building symmetry into the architecture can help to dramatically reduce sample complexity. The question is, how can we exploit this permutation invariant structure in the architecture of a neural network?

[Zaheer et al. \(2017\)](#) propose the following neural network architecture to deal with this problem.

$$f(x_1, x_2, \dots, x_M) = \rho \left(\sum_{m=1, \dots, M} \phi(x_m) \right)$$

where $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$ and $\rho : \mathbb{R}^d \rightarrow \mathcal{Y}$ are unrestricted MLPs.

One can think of this architecture as a two-piece neural network. The first piece maps the inputs into a dimension d . After applying a sum, which is of course permutation invariant, the second piece then maps the dimension d input into the desired output.

The following theorem from their paper justifies their architecture, though the convergence rate is not stated.

Theorem 1. *A function $f(X)$ operating on a set X having elements from a countable universe is invariant to permutation of instances in X iff it can be decomposed in the form $\rho \left(\sum_{x \in X} \phi(x) \right)$ for suitable functions ϕ and ρ .*

2 Simulations

Here are some simple simulations. Each observation is a set $X = \{x_1, \dots, x_M\}$ of $M = 10$ points in $\mathbb{R}^2$, drawn i.i.d. from $\mathcal{N}(0, \Sigma)$, where Σ is redrawn for every set with correlation $r \sim U(-0.9, 0.9)$ and marginal standard deviations $\sigma_1, \sigma_2 \sim U(0.5, 1.5)$.

The regression target is the sample covariance of the two coordinates,

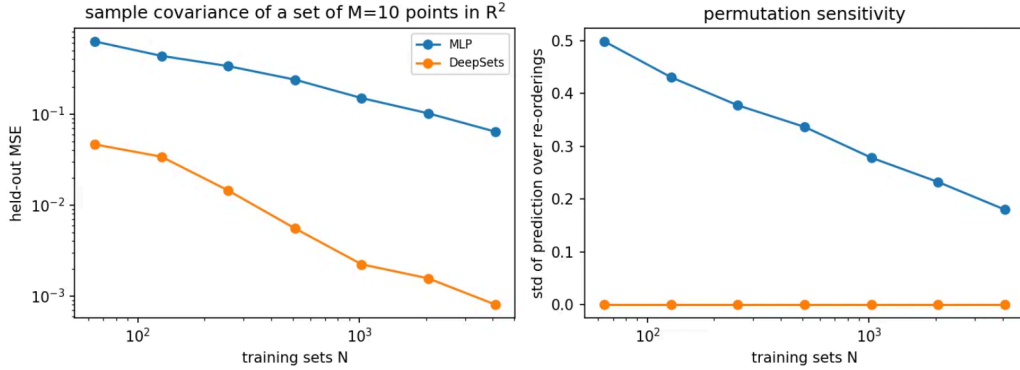
$$y = \frac{1}{M} \sum_{m=1}^M (x_{m1} - \bar{x}_1)(x_{m2} - \bar{x}_2), \quad \bar{x}_j = \frac{1}{M} \sum_{m=1}^M x_{mj},$$

a deterministic, permutation-invariant function of X with $\text{Var}(y) = 0.37$ across sets.

Let's compare two estimators at comparable parameter counts. The unrestricted benchmark is a ReLU MLP applied to the flattened $2M$ -vector, with layer widths 20–64–64–1 (5,569 parameters); it sees the elements in the order they were generated and has no symmetry imposed. The deep sets estimator is

$$\hat{f}(X) = \rho\left(\frac{1}{M} \sum_{m=1}^M \phi(x_m)\right),$$

with $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^{16}$ of widths 2–64–64–16 and $\rho : \mathbb{R}^{16} \rightarrow \mathbb{R}$ of widths 16–64–1, both ReLU (6,545 parameters).



N	MLP			Deep sets		
	MSE		Perm. s.d.	MSE		Perm. s.d.
64	0.6275	(0.0477)	0.499	0.0463	(0.0035)	0.000
128	0.4353	(0.0249)	0.431	0.0339	(0.0002)	0.000
256	0.3360	(0.0321)	0.378	0.0144	(0.0016)	0.000
512	0.2386	(0.0145)	0.337	0.0055	(0.0005)	0.000
1024	0.1497	(0.0073)	0.278	0.0022	(0.0001)	0.000
2048	0.1017	(0.0026)	0.232	0.0016	(0.0002)	0.000
4096	0.0641	(0.0023)	0.180	0.0008	(0.0000)	0.000

Table 1: Held-out performance, mean over three seeds; standard errors across seeds in parentheses.

References

ZAHEER, M., S. KOTTUR, S. RAVANBAKSH, B. POCZOS, R. R. SALAKHUTDINOV, AND A. SMOLA (2017): “Deep Sets,” in *Advances in Neural Information Processing Systems*, ed. by I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Curran Associates, Inc., vol. 30.